MUHAMMAD GEMILANG RAMADHAN

LinkedIn | +62 813-7393-0922 | Portfolio | gemilangramadhan.work@gmail.com | GitHub

Summary

Electrical Engineering graduate with experience in autonomous systems, path planning, machine learning, and software development. Skilled in developing algorithmic solutions, building data-driven applications, and communicating complex technical concepts clearly.

Education

Sepuluh Nopember Institute of Technology — **Surabaya, Indonesia**
Bachelor's degree in electrical engineering, GPA 3.32/4.00 — **Graduating in Oct 2026**

Xaverius 1 Palembang Senior High School — **Palembang, Indonesia**
Science major — **2018-2021**

Work Experience

Kulkas Babe - Surabaya, Indonesia — **Jun 2025 - Aug 2025**
E-Commerce Front-End Web Developer

- Served as the sole front-end developer in a two-person team, building a responsive e-commerce interface using Laravel Blade, Alpine.js, Tailwind CSS, and JavaScript.
- Integrated Midtrans payment gateway for QRIS transactions and Google Maps API for distance-based delivery fee calculation.
- Collaborated with the back-end developer to build maintainable components, standardize development patterns, and conduct structured code reviews.

Anargya ITS EV Team - Surabaya, Indonesia — **Dec 2023 - Nov 2024**
Web Developer & Designer

- Developed and launched the team's public website by translating Figma designs into responsive and accessible Bootstrap pages.
- Created a reusable visual system covering typography, spacing, colors, and digital assets to maintain brand consistency across web and social media platforms.

Projects

USV Path Planning Using Lazy Theta*, SAPF, and DDPG - Surabaya, Indonesia — **Jan 2026 - Jul 2026**
Undergraduate Thesis

- Created thesis called "Implementation of Deep Deterministic Policy Gradient as a Parameter Tuner for Lazy Theta* and Safe Artificial Potential Field in Path Planning for the LSS-01 USV".
- Applied DDPG (reinforcement learning based algorithm) as an adaptive parameter tuner for both algorithms, adjusting heuristic and clearance weights in Lazy Theta*, as well as attractive gain, repulsive gain, and safe distance in SAPF.
- Achieved a simulated path length of 50.3 m and a travel time of 37.8 s using Lazy Theta*-DDPG, outperforming the best fixed-parameter configuration.
- Evaluated the system through simulation and field implementation on the USV LSS-01 platform at Kolam InfinITS ITS.

JejakBatik (Capstone project, team of 5) - Indonesia — **Nov 2024 - Dec 2024**
Machine Learning Engineer

- Architected the end-to-end ML pipeline (data prep, augmentation, training, evaluation) and designed an EfficientNetB0 classifier, achieving 85% pattern-recognition accuracy.

- Implemented real-time image processing & classification with TensorFlow (and TensorFlow.js for mobile preparation) + OpenCV, integrating the model with the Android app for on-device inference.
- Tuned preprocessing and caching to keep inference responsive.

Skills

- **Web Development:** React, JavaScript, Tailwind, Bootstrap, Vite, Node.js, Express
- **Programming & Data:** Python, SQL
- **Machine Learning:** TensorFlow, PyTorch, OpenCV
- **Microsoft:** Excel, Word
- **Tools & Workflow:** Git, GitHub, Jira, Notion
- **Languages:** Indonesian (native), English (professional working proficiency)

Certifications & Achievement

Overall Event Winner, 8th Annual FSEV Concept Challenge, PI-EV **2024**

Contributed as a member of the Anargya ITS EV Team in an electric vehicle concept competition.

Bangkit Academy - Machine Learning Path **Sept 2024 - Dec 2024**

Completed intensive training in machine learning, TensorFlow, supervised and unsupervised learning, OpenCV-based image processing, and pattern recognition.

MySkill Microsoft Excel & SQL for Data Analysis **2023**

Completed training in pivot tables, Power Pivot, SQL fundamentals, basic data visualization, data querying, table manipulation, and basic analysis.